Electrical Bonding and Grounding of Aircraft Structures – General data

1. General

Aircraft electrical bonding is the process by which electrical conductivity is achieved between the component and parts of the aircraft, to ensure electrical continuity of the structure.

Grounding is the electrical connection of a conducting object to primary structure to provide a path for current.

Bonding and grounding connections as specified in these procedures are made in order to ensure the proper operation of the electrical equipment, to suppress the electromagnetic interference resulting from static discharge, electrical shock and lightning strikes and to ensure the aircraft is electrically stable.

2. Types of Bonding Installation

The types of bonding installation required should be specified within the appropriate installations of the PSM 1-64-2. Refer to Table 1 — Type of Bonding Installations. If the type of bonding installation required is not specified within the associated section of the PSM 1-64-2, contact Viking Air Ltd., Technical Support.

Table 1 Type of Bonding Installations

TYPE OF BONDING (AS SPECIFIED IN THE PSM 1-63-2 SECTIONS)	**SURFACE TO BE PREPARED (ELECTRICAL BOND CONTACT AREA)**
Bonding through Direct Contact	refer to Figure 1
Bonding through Attachment Hardware	Nuts/Bolts refer to Figure 2 Screws refer to Figure 3 Rivets Inner surface of rivet hole
Bonding of Connectors	refer to Figure 4
Bonding through Bonding Jumpers (e.g., ground studs)	refer to Figure 5
Bonding of Conductive Plumbing Line Fittings which Pass through the Structure	refer to Figure 6

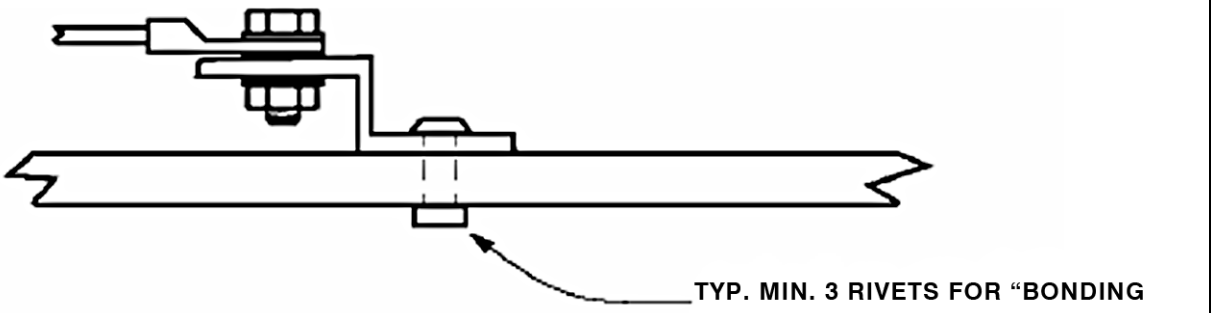
Note Electrically bond externally mounted equipment (i.e. antennas, light housings, data sensors, static wicks, etc.) through direct contact.

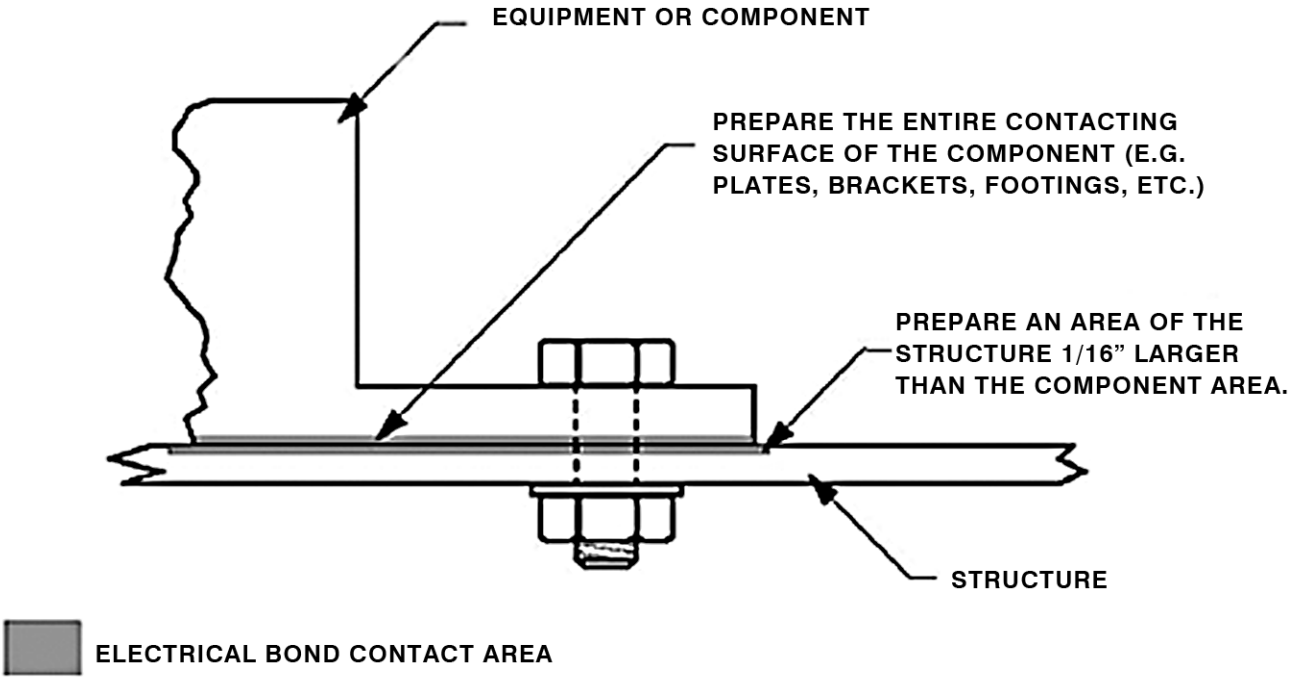
Note After bonding, touch-up the bond points according to Para 10. and seal the bonding and grounding hardware according to Para 11.

Table 1 Type of Bonding Installations (continued)

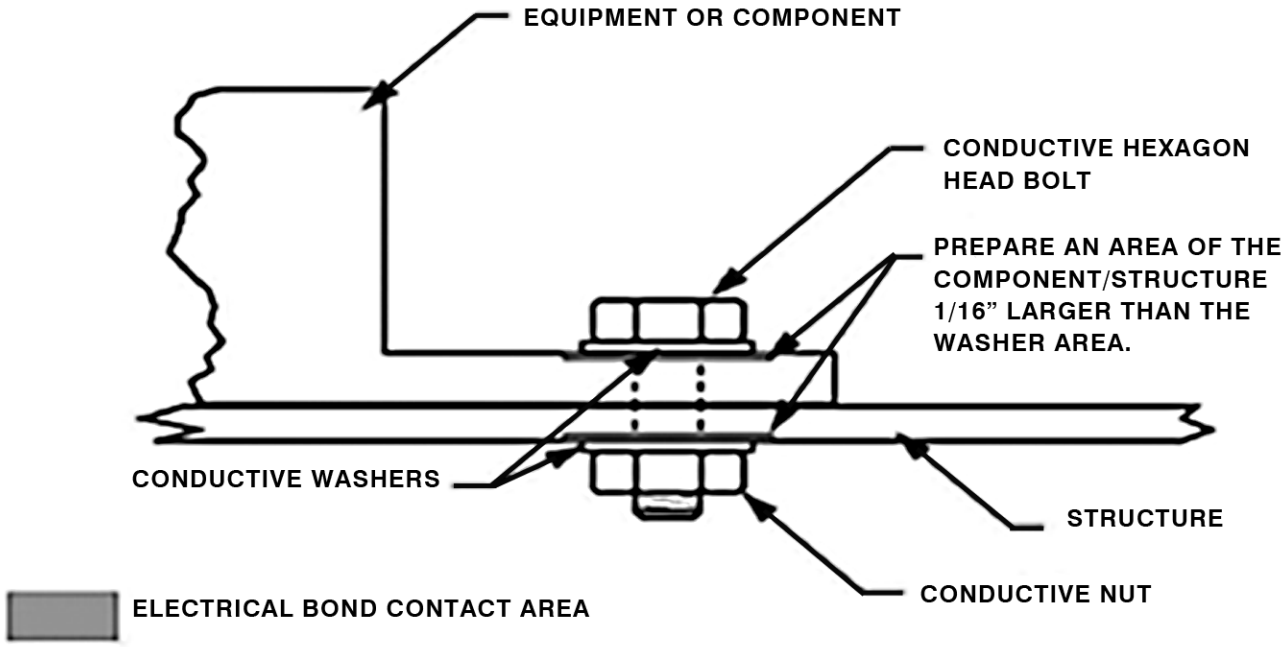
Note "Bonding through Attachment Hardware" for riveted joints must be with a minimum of three rivets if the rivet holes are bare of all insulating finishes, possesses an alodine finish and have been driven (i.e., are not pull type). However, there are high current applications in which more than three rivets are required or where rivets are not acceptable. Specific section of PSM 1-64-2 will specify.

Note In some applications a combination of bonding types is required and will be specified. For example, if a bonding jumper is terminated at a "Z" bracket, as shown below, the instructions for the jumper would be "Bonding through Bonding Jumper" while the instructions for the bracket would be "Bonding through Attachment Hardware" or "Bonding through Direct Contact". Typ. min. 3 rivets for "Bonding through Attachment Hardware".

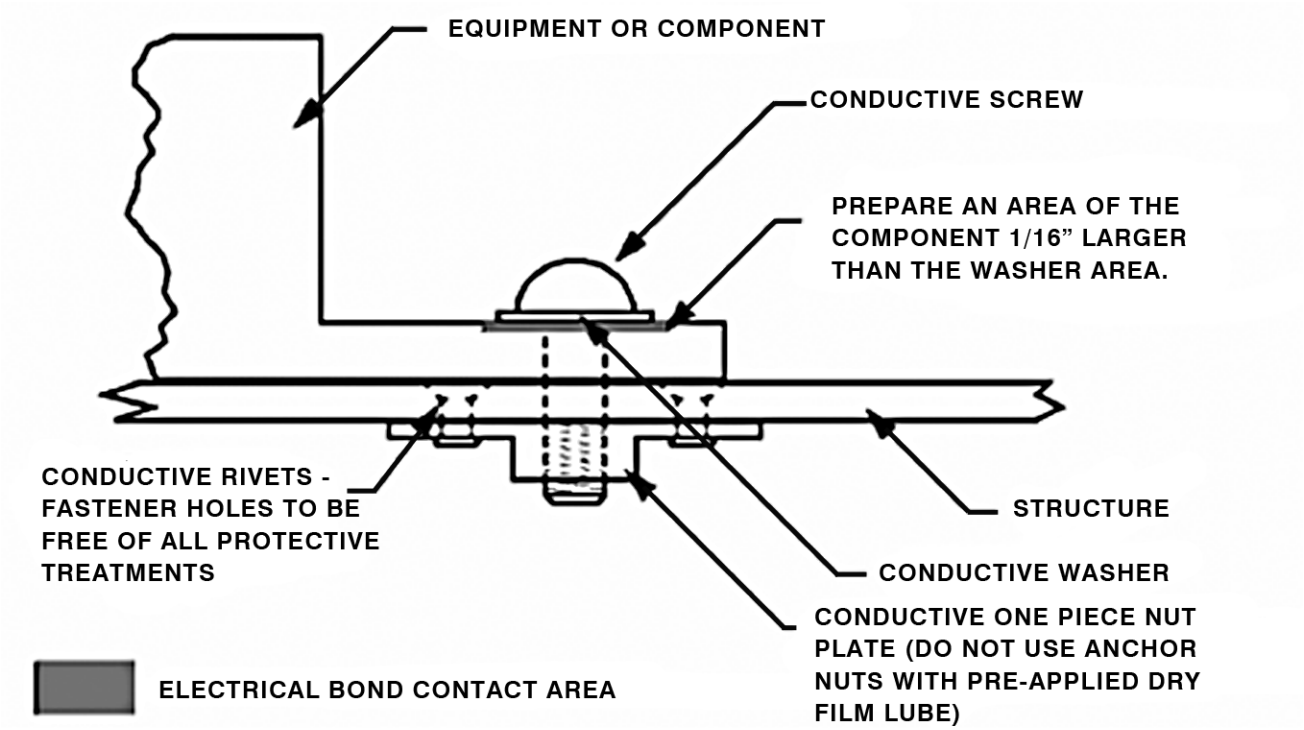




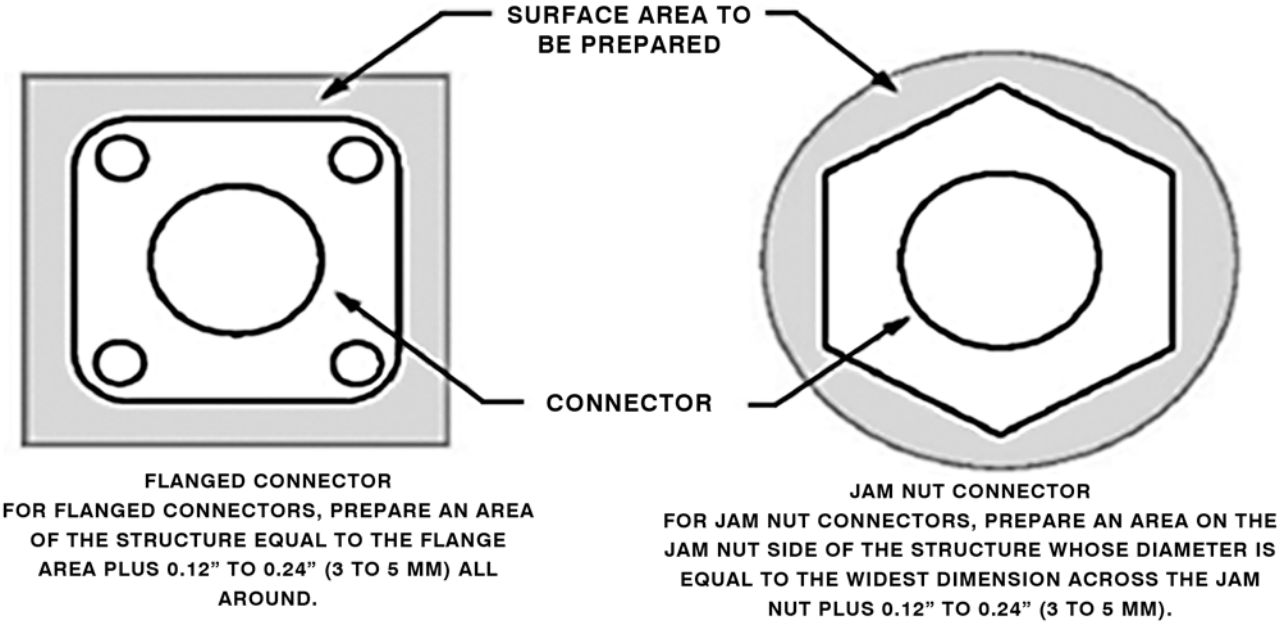
Surfaces to be Prepared for Bonding
Figure 1



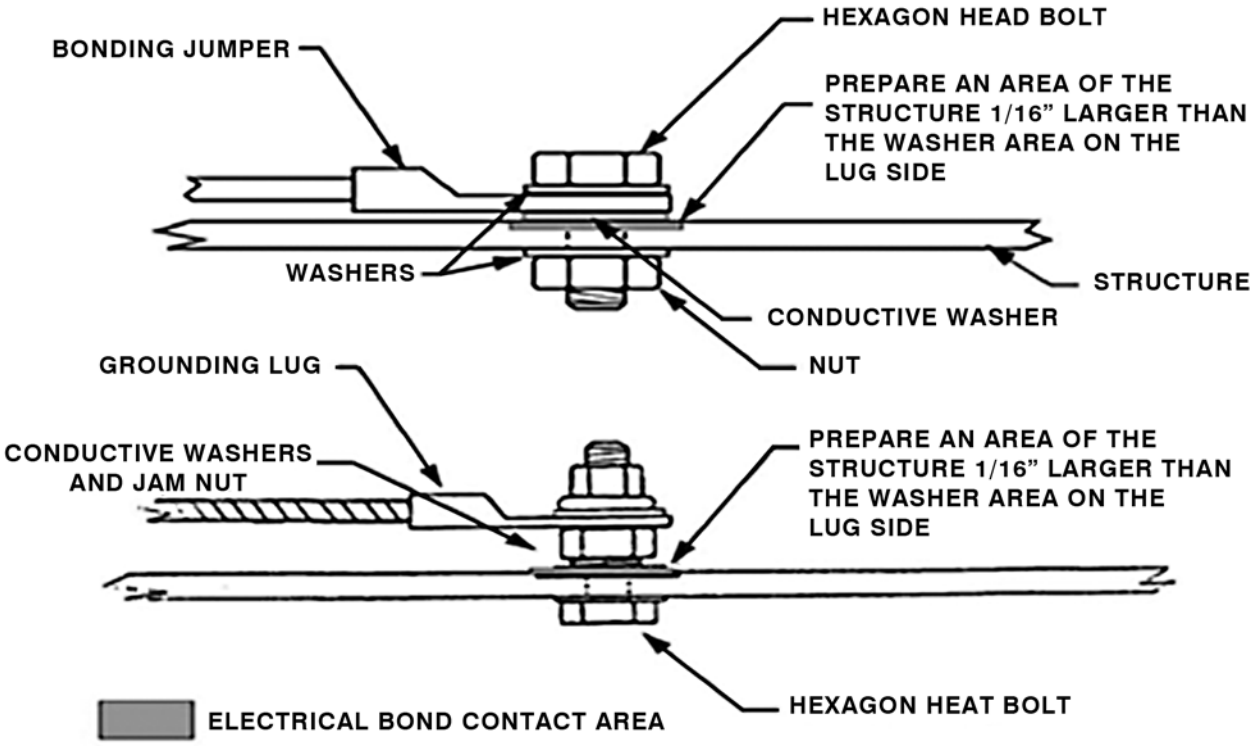
Surfaces to be Prepared with Conductive Nuts and Bolts
Figure 2



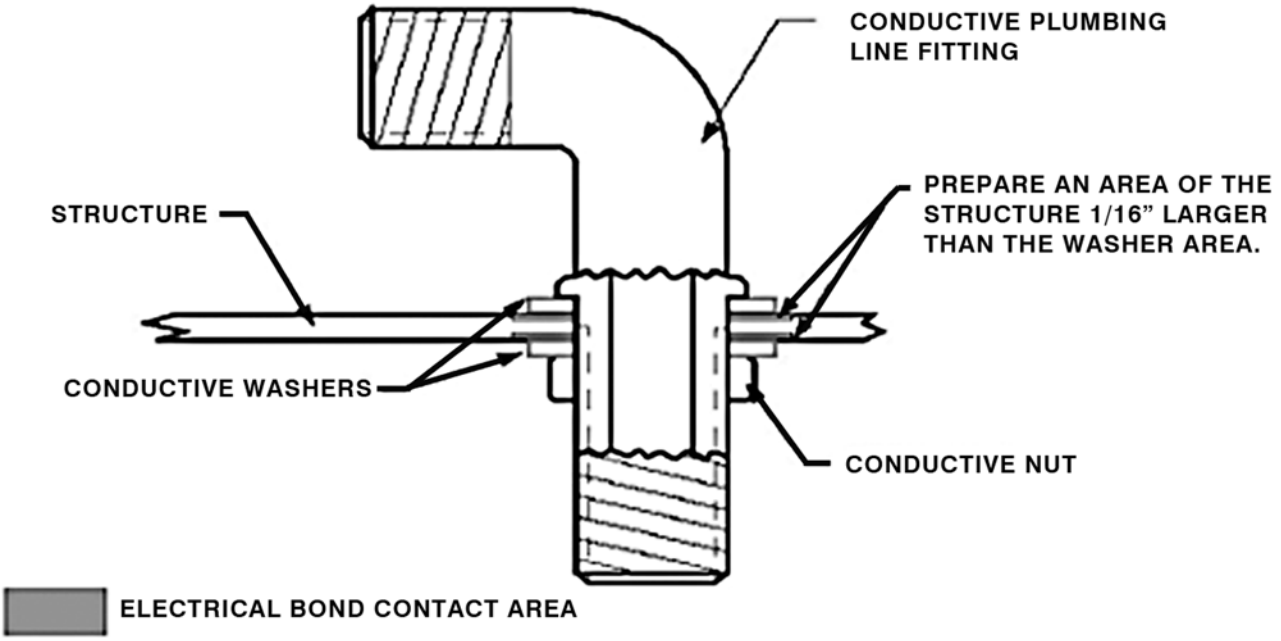
Surfaces to be Prepared for Bonding with Conductive Screws and Anchor Nuts
Figure 3



Surfaces to be Prepared for Bonding of Electrical Connectors
Figure 4



Surfaces to be Prepared for Bonding through Bonding Jumpers (Typical)
Figure 5



Electrical Bonding of Plumbing Lines Fittings to the Structure
Figure 6

3. Requirements

A. Check that all bonding point surfaces were thoroughly cleaned before the installation of clamps and bonding jumpers. This ensures a satisfactory bond.

B. Check pipe lugs for cracks, bonding terminals for insecure wire braiding, and clamps and connections for looseness.

C. Ensure that all bond connections are tightened to the torque value specified in Table 2 — Bonding Hardware Torque Values.

D. Ensure that the method and materials used for bonding and ground connections are in conformance with the standards and procedures in 20-80-00, Installation of Aircraft Antennas – General data.

E. Ensure that bonding jumpers are installed so that the wire does not come into contact with surrounding structure, equipment, fuel pipes, or other bonding jumpers (refer to 20-75-01, Application of Bonding Jumpers – General data, Figure 2). If contact is unavoidable, use a CSP 476 bonding jumper (complete with insulating sleeve). If there is a jumper other than the CSP 476, contact Viking Air Ltd.

F. For cable harnesses with individually shielded wires but without overbraid shields, verify proper shield termination according to PSM 1-64-2W, as applicable, before the backshell is closed.

G. Ensure that touch-up primer and, if applicable, paint top-coat, has been applied to any aircraft structure remaining exposed after installation of all bonding and grounding hardware.

4. Installation of Bonding Hardware and Ground Studs

A. Bonding Hardware

Install bonding hardware and connections according to the specified type of bonding installation. If the type of bonding installation required is not specified within the associated section of the PSM 1-64-2, contact Viking Air Ltd.

When installing the bonding hardware, including jumpers, check that all of the nuts or attachment screws are fully tightened.

Table 2 Bonding Hardware Torque Values

SCREW SIZE	CONVENTIONAL AND BLIND TYPE STUDS (INCH POUNDS)	PEM STUDS (INCH POUNDS)
#6	N/A	5 - 9
#8	15 - 20	15 - 18
#10	25 - 30	25 - 30
1/4	40 - 45	N/A
5/16	80 - 85	N/A
3/8	110 - 125	N/A
1/2	140 - 155	N/A

B. Ground Studs

1) Install all blind type bonding and grounding terminal studs in accordance with manufactures recommendation.

2) Install all PEM studs for use as bonding and grounding terminal studs in accordance with manufactures recommendation.

3) Torque all ground stud jam nuts and securing nuts to the values shown in Table 2 — Bonding Hardware Torque Values.

5. Installation of MS90298 Fuel Nozzle Grounding Receptacles

A. Prepare the inner surface if the skin around the hole is drilled as specified in Para 6.

B. Install the receptacle, including the inner and outer washers, and tighten the nut securely so that the receptacle does not rotate and the inner washer is firmly in contact with the skin.

C. Carry out the bonding resistance measurement test in accordance to Para 8.C.

D. Stake the receptacle nut at 3 points, approximately 120° apart, and seal the interior hardware and exterior hardware as specified in Para 11.

6. Surfaces to be Prepared for Electrical Bonding

A. Except as noted below, the electrical bond contact area refers to an area approximately 1/16” larger all around than the area of the component being bonded or, in the case of loop type clamps on tubing, approximately 50% greater than the clamp contact area.

Note Voi-Shan E-M grounding studs utilize the fastener hole to establish the electrical conductivity. Therefore, for Voi-Shan E-M grounding studs, the electrical bond contact area is limited to the fastener hole.

B. If surface is covered with a high resistance film then:

1) Strip the high resistance film from the electrical bond contact area as per Para 7.

Note Strip high resistance films on the grounding lug side to provide an acceptable electrical bond contact area.

Note After stripping any high resistance films, test the electrical bond contact area for consistent electrical conductivity and then chemical conversion coat, Class 3, within 8 hours of stripping the high resistance films.

2) Test the electrical bond contact area for consistent electrical conductivity according to Para 8.C.

3) If the electrical conductivity is not consistent over the electrical bond contact area then repeat Para 6.B.1) and Para 6.B.2) as required.

4) Touch up the electrical bond contact area with chemical conversion coating Class 3 as per 20-30-02, Chemical Conversion Coatings (Alodine) – Manual Application).

5) Solvent clean the electrical bond contact area as per Para 7.

6) Install component as per the applicable section PSM 1-64-2.

C. If the surface is not covered with a high resistance film then:

1) Solvent clean the electrical bond contact area as per Para 7.

2) Install component as per the applicable section in PSM 1-64-2.

D. The following finishes and surface conditions are satisfactory for bonding connections and do not need to be removed. Solvent clean these finishes according to Para 7. immediately before installation:

1) Plated surfaces

2) Graphite (powder, flake or stick)

3) Chrome pickle or selenious acid treatment on magnesium alloys

4) Conductive Chemical conversion coating (e.g. MIL–C–5541 Class 3)

E. If specified in the associated sections of PSM 1-64-2, coating the shank of a conductive bolt with F13, ensure that no F13 gets under the bolt head.

7. Surface Preparation

A. Removal of High Resistance Films

1) Materials and Equipment

- SD 8383 spot facing tool
- Abrasive paper, aluminium oxide, 180 – 240 grit
- Tack rag

2) Removal of High Resistance Films

Except as noted below, mechanically remove oxides and anodic films, dichromate treatment, paint, primer, etc. Using either an SD 8383 spot facing tool or abrasive paper. Keep the amount of metal removed during mechanical cleaning to a minimum.

a) On structure other than aluminium alloy, it is also acceptable to use a stainless steel wire brush to mechanically remove oxides and anodic films, dichromate treatment, paint, primer, etc. Do not use stainless steel wire brushes on aluminum alloys.

b) On aircraft fluid lines which have been F19 primed (without bond point masking), locally strip the primer from the bond contact area using chemical paint stripper according to manufactures recommendations, use masking tape to limit the extent of stripping.

c) If mechanical stripping is impractical, strip paints, dyes or stains from bond contact areas chemically according to manufactures recommendations. Use masking tape to limit the extent of stripping. When removing non-conductive coatings, do not use the same abrasive tool on different kinds of metal (i.e. don't use a toll on aluminum parts that have been used on steel parts).

3) After mechanically abrading the surface, use a tack rag to remove any dust or adhering particles.

4) Treat the bond contact areas of magnesium alloy parts which have been mechanically cleaned with 10% selenious acid solution according to manufactures recommendations.

5) Thoroughly clean all bond contact surfaces (including terminal lugs of bonding jumpers) by solvent wiping them according to Para 7.B immediately before installation.

B. Manual Solvent Cleaning

Note Solvent clean area about 50% greater than sealant contact area.

1) Materials and Equipment

- Clean Lint-free wiping clothes
- Protective Gloves (e.g. DSC 422–1 and DSC 422–2)
- Approved Respirator
- Approved Safety glasses

2) Manual Solvent Cleaning

Note Solvent clean an area about 50% than sealant contact area.

WARNING KEEP SOLVENT AWAY FROM FIRE AND OTHER SOURCES OF IGNITION. PROVIDE SUFFICIENT VENTILATION WHEN WORKING IN CONFINED SPACES. AVOID SKIN CONTACT WITH SOLVENT.

3) Apply small amount of solvent to a clean wiping cloth. Refer to 20-40-01, Silicone Adhesives – General Data, Table 1 for the solvent appropriate for the surface to be cleaned. For solvents which evaporate quickly, apply more solvent for solvents which evaporate slowly.

4) Wipe the area with a solvent dampened wiping cloth.

a) When cleaning bonded surfaces with solvent saturated wiping cloth, thoroughly scrub the bonding surfaces. Wipe dry with a cloth before solvent evaporates.

b) Keep solvent cloth saturated and turn frequently to avoid reusing a dirty area.

5) Wipe dry with another clean wiping cloth before the solvent evaporates.

Note It is important to wipe the surface immediately after cleaning because if the solvent evaporates before it is wiped up, the oil, and grease residue will remain in the surface. Although the DS 108 solvent has a slow evaporation rate, wipe dry as soon as possible.

6) Check for a water free surface.

7) If more cleaning is needed, apply fresh solvent to a clean portion of the wiping cloth and repeat Para 7.B.1) through to Para 7.B.6).

8. Resistance Measurement

A. General

1) If possible, take resistance measurements or mask areas where resistance measurements are to be taken before applying non-conductive coatings.

2) If non-conductive coatings have been applied to the area where the resistance measurement is to be taken, remove an area of coating sufficient to accommodate the ohmmeter probe according to Para 7.A before taking the measurement.

3) If the protective coating has been removed from a part or an area of the part was masked to accomplish resistance measurement, touch up the area as follows after taking the measurement:

B. Equipment

- Approved Safety glasses
- Multimeter, 1 megohm range.
- 500 volt Insulation Resistance Tester (e.g., Biddle Instruments P/N 210801–2)
- Spot facing tool (e.g., SD 8383)

Note Any of the following ohmmeters may be used, provided hardware and training is available:

- BCD M1 Milliohm meter
- Biddle Instruments digital low resistance ohmmeter, DRLO #247001
- SD 8778–1 10 amp constant current ohmmeter test set
- Ducter Instruments/Megger Instruments BT51 digital low resistance ohmmeter
- Ducter D203 Micro-Ohmmeter by Megger Instruments
- HP 4328A Milliohm meter
- HIOKI 3220 or 3540 Milliohm meter
- Eaton Corp. model 584 resistance tester
- Eutron S.A.S. Micro-Ohmmeter BVM 3–10
- AOIP Measures OM 21–1 Ohmmeter
- AEMC Micro-Ohmmeter, Model 5600

C. Testing of Stripped or Masked Electrical Bond Contact Areas

1) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing both probes of a BCD M1 Milliohm meter on the stripped/masked surface and measuring the resistance. The resistance between the two points must be no more than 2.5 milliohms unless stated otherwise.

Note It is acceptable to use one of the ohmmeters specified in Para 7.A as an alternative to the BCD M1 Milliohm meter.

2) Check the electrical conductivity between several random pairs of points within the electrical bond contact area to ensure consistency.

3) If the electrical conductivity is not consistent over the electrical bond contact area, remove the non-conductive coating as specified in Para 7. and re-test the area.

9. Electrical Component Bonding

All Rack Mount/Remote Mount units shall be electrically bonded to the airframe. This is accomplished by ensuring that the mating surfaces between the LRU mounting tray (or LRU mounting feet if a tray is not used) provides a low impedance electrical path. The mating surfaces must be free of all paint and other non-conductive elements and should be burnished to ensure a good bond.

All units must be properly grounded to the airframe ground. Resistance measurements to be made between the unit mounting tray or unit mounting feet if a tray is not used (or other acceptable/known unit ground) and the applicable mounting surface. Maximum resistance per each electrical bonding interface is 2.5 milliohms not to exceed 10 milliohms from unit to primary structure.

Panel/Pedestal Mount Units cannot be readily grounded to the airframe via the panel mounting screws of DZUS fasteners. Grounding of these units is accomplished by using the shield grounding braid or bonding jumper bonded to the airframe.

10. Touch-up of Bond Points and Resistance Measurements Points

A. Except as noted below, if the protective coating has been removed from a part or an area of the part that was masked to accomplish electrical bonding or resistance measurement, solvent clean the affected area according to Para 7. and touch-up the area with F19 primer applied by brush within 24 hours of installing the bonding hardware:

Note Touch-up integral fuel tanks with F21 Type II primer applied by brush.

Bond points which are to be potted with sealant as specified in Para 11. do not require primer touch-up.

B. Ensure that primer refinishing provides complete coverage of the affected area and cover an area approximately twice that of the area cleaned for bonding/resistance measurement.

C. If the affected area requires paint touch-up, brush coat the area with the top-coat paint system matching the existing.

D. Touch-up any unused bonding points provided at the ends of primed fluid lines by:

1) Solvent clean the affected area according to Para 7.

2) Touch-up the area with F19 primer applied by brush within 24 hours of installing the bonding hardware.

11. Sealing of Bonding and Grounding Hardware

Refer to Figure 1 and Figure 2 for the typical areas to be sealed on the various types of bonding assemblies. For clamps which secure/support overbraid harness and provide electrical bonding and grounding, pot the clamp attachment point to the structure; however, it is not necessary to pot or seal in the area where the clamp contacts overbraid the harness.

A. Materials

- Sealant spatulas (e.g. SD9164, Mk1, Mk2, Mk3 and Mk4)
- PR1776M Sealant
- Humiseal 1B15 Sealant if no sealant has been specified
- Neozapon red dye only if Humiseal 1B15 is used
- Humiseal 1B31LOC or Humiseal 1B31EPA

Note 1 Humiseal 1B31LOC and Humiseal 1B31EPA contain no MEK, and for this reason are preferred over Humiseal 1B15 sealant. Humiseal 1B31LOC and Humiseal 1B31EPA do not need to be dyed red, they can be inspected using UV lighting tools.

2 Before use, Humiseal 1B15 sealant must be dyed red by mixing in 4 - 6 grams of BASF Corp. 335 Neozapon red dye per U.S. quart of sealant. Humiseal 1B15 sealant can be purchased with red dye pre mixed or, sealant and dye separately, to be mixed prior to use. The purposes of dying the Humiseal 1B15 sealant is to make it visible for inspection purposes, so some variation in the shade and intensity of red colour is acceptable.

B. Equipment

- Protective Gloves (e.g. DSC 422–1 and DSC 422–2)
- Approved Safety glasses

C. Preparation

1) Solvent clean an area about 50% greater than sealant contact area, in accordance to Para 7.

2) When installing the bonding hardware, including bonding jumpers, check that all of the nuts or attachment screws are fully tightened.

3) Pot the electrical bonding or grounding attachment points and hardware on the fuselage as well as those inside the fuel tanks with sealant. If the sealant is not specified then pot as follows:

a) Inside the fuselage (baggage compartments, electrical bays, avionics bays, electrical box assemblies, cockpit, instrument panel, cabin, tailcone) pot with either PR1776M or Humiseal sealant (Humiseal 1B31LOC, Humiseal 1B31EPA or Humiseal 1B15).

CAUTION DO NOT USE ANY TYPE OF HUMISEAL SEALANT IN AREAS SUBJECT TO HIGH FREQUENCY EXPOSURE TO SKYDROL HYDRAULIC FLUID.

b) Inside fuel tanks, pot with PR1776M.

CAUTION DO NOT USE ANY TYPE OF HUMISEAL SEALANT INSIDE FUEL TANKS.

c) For exterior and high moisture areas (landing gear, empennage, hydraulic bay, fuel gallery, fuel filling provisions, windscreens, wire antennas, inside radome), pot with PR1776M sealant.

d) Unless specified, it is acceptable to seal electrical bonding attachment points and hardware that are part of an electrical box or panel with either PR1776M or Humiseal.

4) If Humiseal sealant is used:

a) Before using Humiseal 1B15 sealant, dye the sealant red per Para 11.A, Note 2.

b) Before sealing with Humiseal, clean the area using LPS CFC Free cleaner. Gentle air drying with compressed air is permitted.

c) Apply Humiseal coating using a paint brush of a size suitable for the area to be coated. Humiseal 1B15 may be brushed with a small addition of Humiseal Thinner 503. The consistent application of the film depends on component density and the operator's technique.

5) If PR1776M sealant is used prepare in accordance to manufacturer specifications.

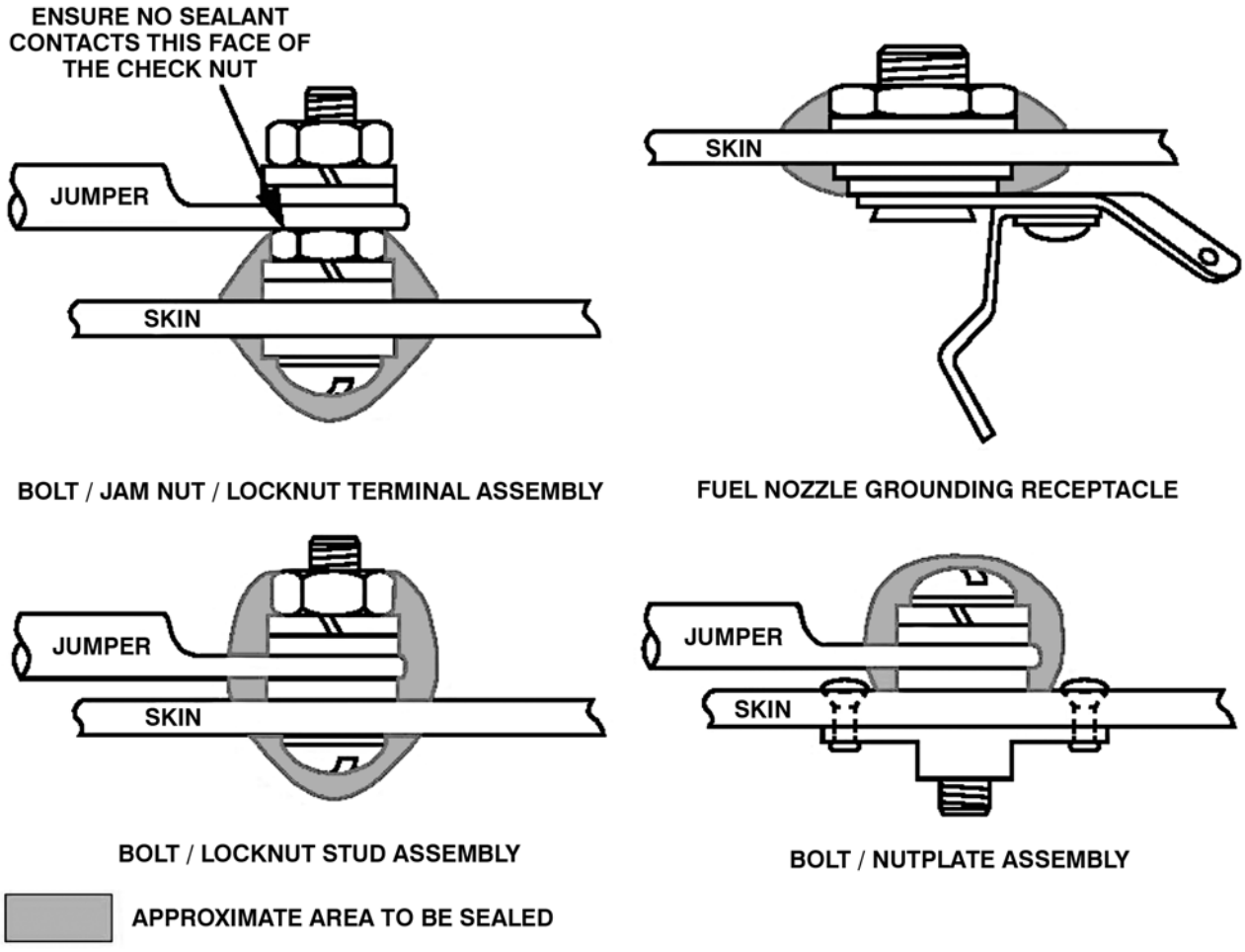
D. Application of Sealant

1) Apply sealant to the electrical bonded assemblies using the procedures listed in 20-50-01, Weather Sealing – General Data. For the final application of sealant refer to the electrical bonded assemblies as shown in Figure 7 and Figure 8 for areas NOT exposed to high levels of moisture, and Figure 9 and Figure 10 for areas exposed to high levels of moisture.

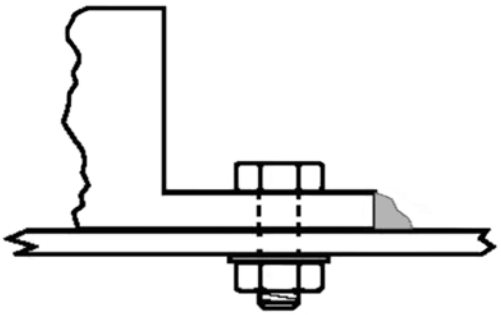
2) When installing bonding hardware on hardware metallic tubes, including clamps, bonding jumpers and hardware, seal the clamp around the circumference of the tube, screw, nut, washers and the ring portion of the terminal.

3) Extend the sealant approximately 1/8” beyond the area stripped of protective coating.

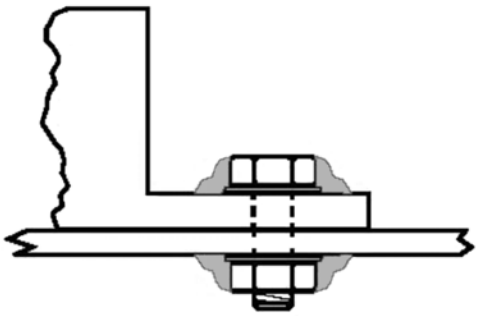
4) For M81714/27 stud mounted ground modules installed outside the fuselage and in high moisture areas, seal the stud with PR1776M sealant. Touch up bare metal using F19 primer.



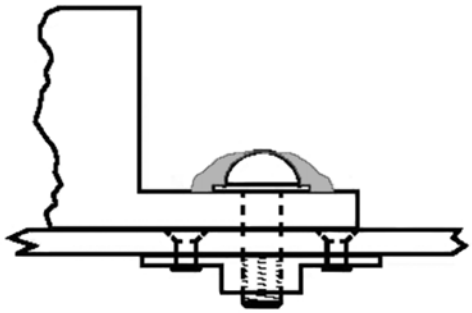
Sealing of Electrical Bonding Assemblies – Indirect Bonding (NOT exposed to high levels of moisture)
Figure 7



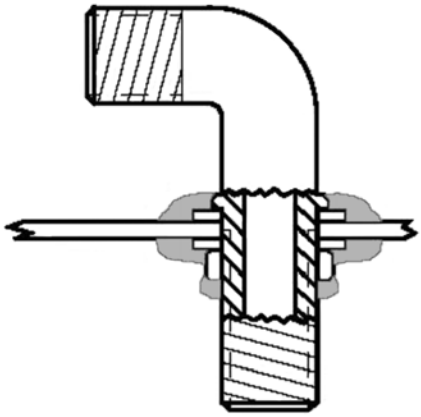
DIRECT CONTACT BONDING



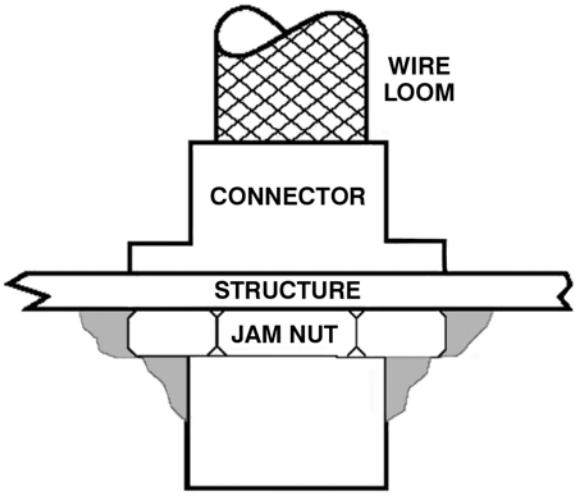
BONDING WITH CONDUCTIVE NUTS AND BOLTS



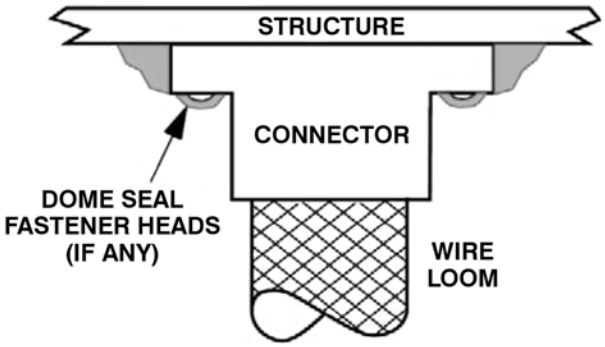
BONDING WITH CONDUCTIVE SCREWS



BONDING OF PLUMBING LINE FITTINGS TO THE STRUCTURE



BONDING OF JAM NUT ELECTRICAL CONNECTORS

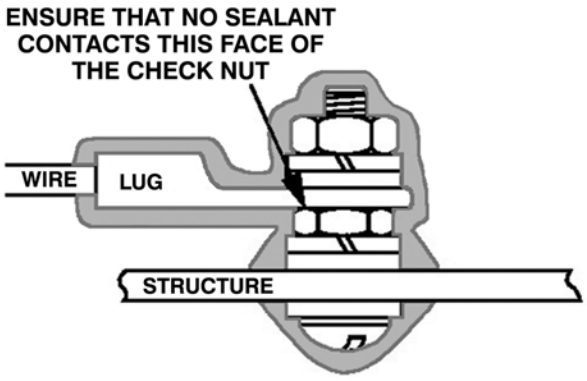


BONDING OF FLANGED ELECTRICAL CONNECTORS

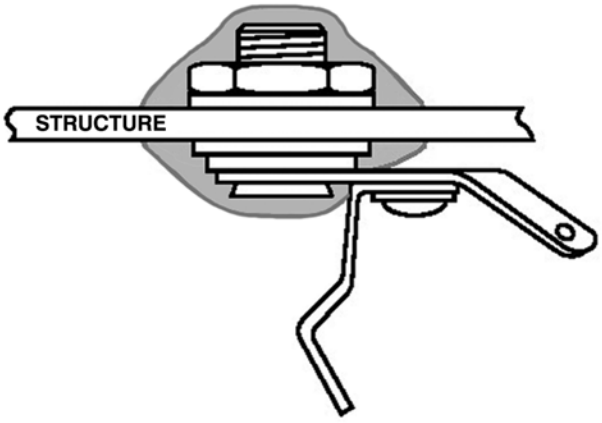


APPROXIMATE AREA TO BE SEALED

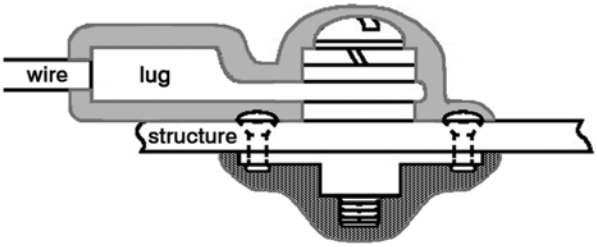
Sealing of Electrical Bonding Assemblies – Direct Bonding (NOT exposed to high levels of moisture)
Figure 8



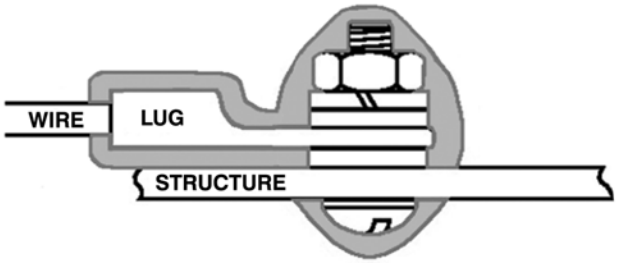
BOLT/JAM NUT/LOCKNUT TERMINAL ASSEMBLY



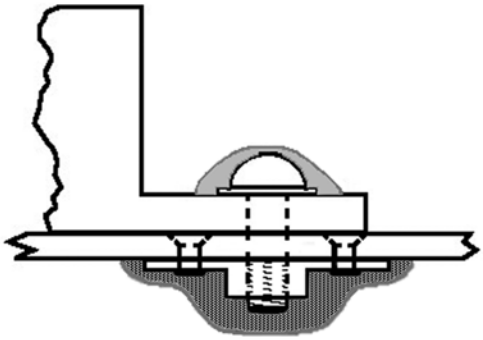
FUEL NOZZLE GROUNDING RECEPTACLE



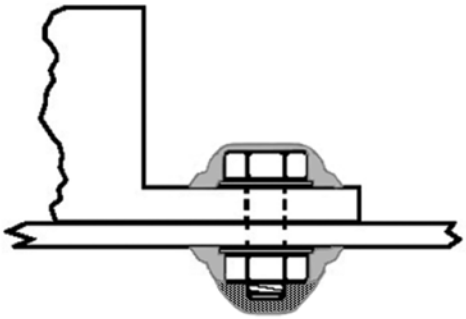
BOLT/NUTPLATE ASSEMBLY



BOLT/LOCKNUT STUD ASSEMBLY



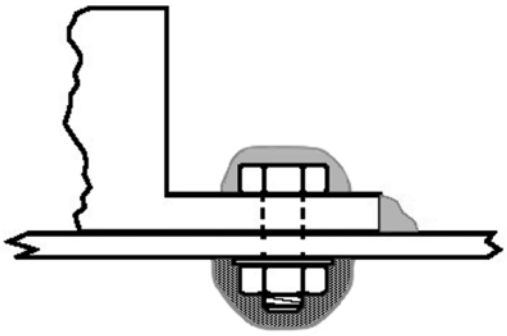
BONDING WITH CONDUCTIVE SCREW/NUTPLATE



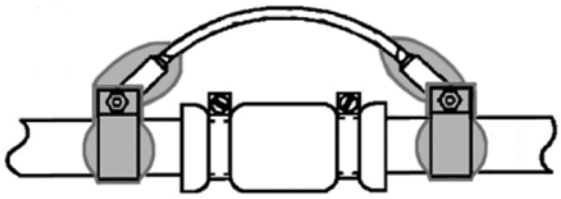
BONDING WITH CONDUCTIVE SCREW/LOCKNUT

NOTE: APPROPRIATE SEALING AREA SEALING IF ACCESS PERMITS

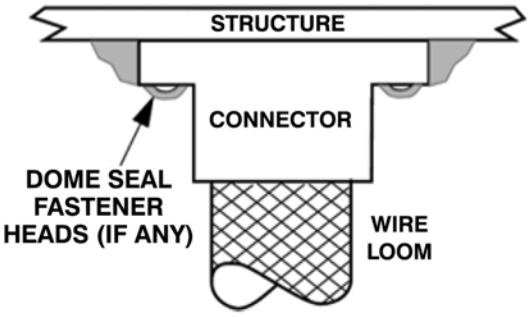
Sealing of Electrical Bonding Assemblies – Indirect Bonding (exposed to high levels of moisture)
Figure 9



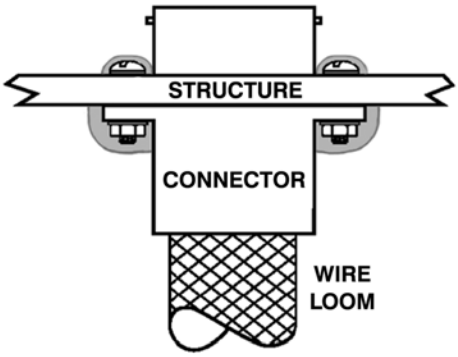
DIRECT CONTACT BONDING



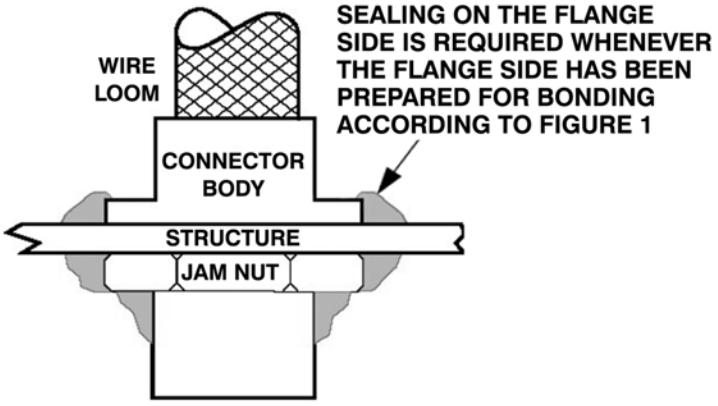
LOOP CLAMP BONDING



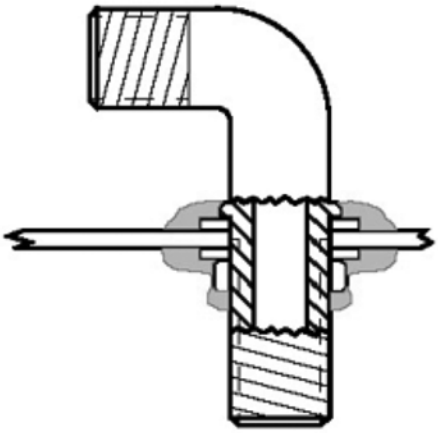
FLANGE CONNECTOR BONDING



FLANGE CONNECTOR BONDING



JAM NUT CONNECTOR BONDING



PLUMBING LINE FITTING BONDING

NOTE: APPROPRIATE AREA TO BE SEALED SEALING IF ACCESS PERMITS

Sealing of Electrical Bonding Assemblies – Direct Bonding (exposed to high levels of moisture)
Figure 10

This Page Intentionally Left Blank